# Tail-GAN: Learning to Simulate Tail Risk Scenarios

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TAiLgan
Learning to simulate tail Risk scenarios

# Market simulator

**What is a market simulator?**

- Denote $\mathbb{P}_r$ as the target distribution of some price dynamics over a given time period. Given some samples $\{p_i\}_{i=1}^N$ from $\mathbb{P}_r$, the **goal** is to construct $\mathbb{P}_G$ such that

$$\mathbb{P}_G \text{ is "similar" to } \mathbb{P}_r \text{ under certain criterion}$$

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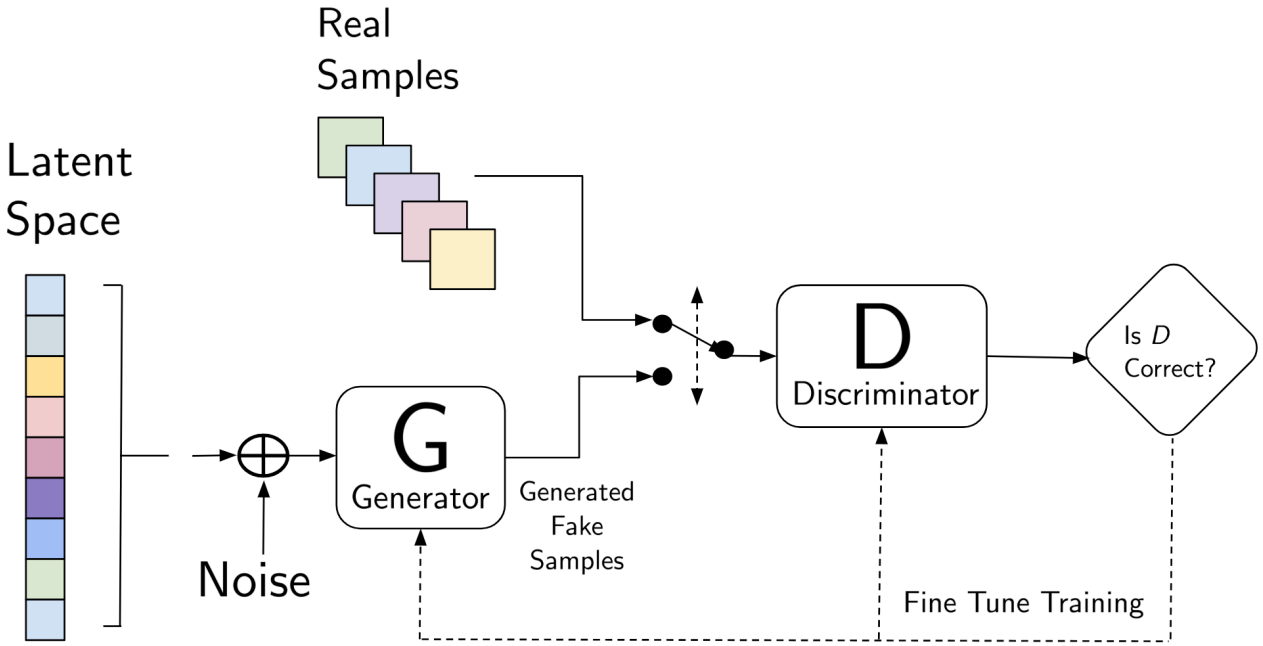
**Why do we need market simulator?**

- Key ingredient of quantitative risk management systems.
- Main application: estimate the loss distributions for various trading strategies and portfolios.
- One focus is the calculation of **risk measures**, such as Value-at-Risk and Expected Shortfall.
- Other applications:
  - back-testing and stress testing; improve risk management; portfolio construction; forecasting
  - data protection

# Classic simulation models in mathematical finance

- **Parametric approach**: time series models such as GARCH models, diffusion and jump processes, stochastic volatility models, and copulas
  - **Pros:** successful in low-dimensional applications involving a small number of homogeneous assets
  - **Cons:** difficult to model the joint dynamics of large or heterogeneous portfolios with assets from different classes $\Rightarrow$lack of scalability + risk of model misspecification
- **Gaussian models** ("Black-Scholes models")
  - Default approach in most high-dimensional applications
  - **Cons:** mismatch with stylized facts observed from financial data (e.g., heavy tails, volatility clustering)

# Generative Adversarial Networks (GAN)




[Goodfellow et al., 2014]

- Robust network used for unsupervised machine learning to build a min-max game between two players: D (the discriminator) and G (the generator)
- Player G tries to fool player D by producing very "natural looking" data from random latent vector $z$
- Player D gets better in-distinguishing between real and generated data

# GAN in finance...

- GANs have been used for building 'market simulators' in finance (e.g. [Koshiyama et al., 2020, Wiese et al., 2020, Buehler et al., 2020, Yoon et al., 2019]) by training them on raw returns.
- Training criterion is cross-entropy or some probability metrics, e.g. Wasserstein distance.
- Final result is a 'black-box' simulator that generates market scenarios.

| Reference | Model category | Architecture | Loss Function | Data |
|---|---|---|---|---|
| Quant-GAN [Wiese et al., 2020] | GAN | TCN | KL divergence | Returns |
| [Buehler et al., 2020] | VAE | NN | $l_2$ norm | Signatures of paths |
| Time-GAN [Yoon et al., 2019] | GAN + Supervised Learning | RNN | Combination of log-likelihood and $l_2$ norm | Time Series |
| [Fu et al., 2019, Koshiyama et al., 2020] | CGAN | NN | KL divergence | Returns |
| SigCWGAN [Ni et al., 2020] | CGAN | Residual NN | $l_2$ norm | Signatures |
| [Li et al., 2020] | CGAN | LSTM | KL divergence | Limit order books |
| PCFGAN [Ni et al 2023] | GAN | NN | Characteristic fn | Time series |
| ExGAN [Bhatia et al., 2020] | CGAN | CNN | KL divergence + Supervised loss | Rainfall images |

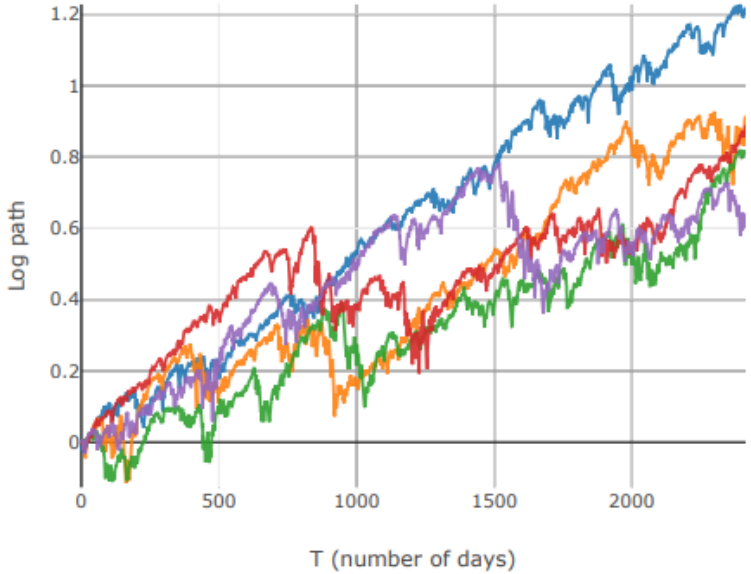
Table 1: Literature review.

# Key questions in model validation

How do we know if the generated scenarios are realistic?

- In the context of images or text, user perception can be easily used for validation
- virtually check is impossible to evaluate the simulated finance scenarios

(a) Image generation of dogs




(b) Financial scenarios: which path is a possible realization of S&P500?

# Validation Criterion

Criterion should be tailored to the down-stream tasks of interest.

Prior works focus on "global properties" (such as Wasserstein distance) of the return distribution:

- Global property **v.s.** Tail behaviour (rare events)?
- Raw return distribution **v.s.** PnL distribution of trading strategies?

# Overview of TAIL-GAN framework

Tail risk and Elicitability:

- Value at Risk (VaR) and Expected Shortfall (ES) are commonly used statistics for measuring the tail risk of a variety of trading strategies
- VaR and ES of a group of trading strategies are incorporated in the loss function by utilizing the joint *elicitability* property

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Discrimination via dynamic trading strategies:

- static portfolios are used to capture cross-asset dependence information
- dynamic strategies are used to capture temporal dependence information in the financial scenarios

# Outline

Motivation and Introduction

Score Functions for Risk Measures

TAIL-GAN: Design and Theoretical Guarantee

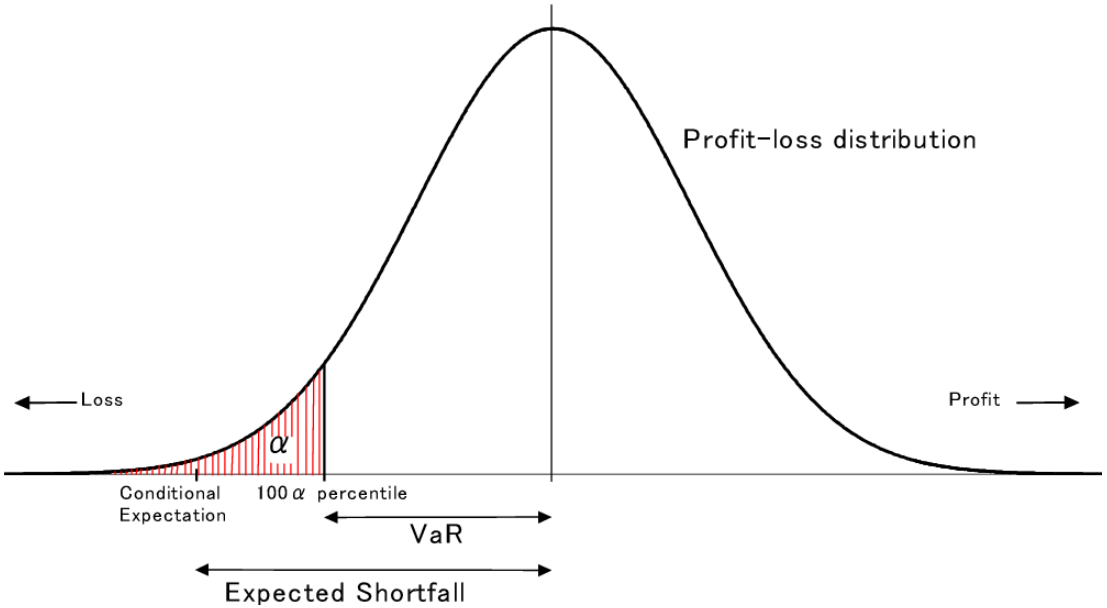
Numerical Performance

# Tail risk measures

## Value-at-Risk (VaR) and Expected Shortfall (ES)

Given a one-dimensional distribution $\mu$ with a finite mean and a confidence level $0 < \alpha < 1$, the $\alpha$-VaR and $\alpha$-ES are defined as

$$\begin{aligned} \mathrm{VaR}_\alpha(\mu) &:= \inf\left\{x \in \mathbb{R} : \mu(X \leq x) \geq \alpha\right\} \\ \mathrm{ES}_\alpha(\mu) &:= \frac{1}{\alpha}\int_0^\alpha \mathrm{VaR}_\beta(\mu)\mathrm{d}\beta \end{aligned}$$



# Elicitability: from statistical estimation to optimization

## Elicitability

A statistical functional $T$ (e.g., mean) is **elicitable** for a set of distributions $\mathcal{F}$ if there is a score function $S(x, y)$ such that

$$T(\mu) = \arg\min_x \int S(x, y)\mu(\mathrm{d}y)$$

for any distribution $\mu \in \mathcal{F}$

**Examples:**

- mean with $S(x, y) = (x - y)^2$
- median with $S(x, y) = |x - y|$
- $\mathrm{ES}_\alpha$ is not elicitable but $(\mathrm{VaR}_\alpha, \mathrm{ES}_\alpha)$ is jointly elicitable
- Spectral risk measure $\Rightarrow$ allows us to incorporate multiple quantile levels

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# Some notations

- Consider the price dynamics of $M$ assets over a time horizon $T$
- The user specifies $K$ trading strategies which may be static or dynamic
- $\Pi^k : \mathbb{R}^{M \times T} \to \mathbb{R}$ represents the $k$-th strategy, with the price scenarios $\mathbf{p} \in \mathbb{R}^{M \times T}$ as the input and the final PnL as the output
- $\Pi^k \# \mathbb{P}_r$ denotes the distribution of the PnL of the k-th strategy $\Pi^k(\boldsymbol{p})$, where $\boldsymbol{p} \sim \mathbb{P}_r$.

# Theoretical version of the discriminator

- $D : \mathcal{P}(\mathbb{R}) \to \mathbb{R}^2$ a mapping parameterized by a neural network, with strategy PnLs as the input and two values as the output
- Goal:

$$\overbrace{D(\underbrace{\Pi^k \# \mathbb{P}_r}_{\text{strategy PnL distribution}})}^{\text{VaR and ES prediction from D}} \approx \Big(\mathrm{VaR}_\alpha\big(\Pi^k \# \mathbb{P}_r\big), \mathrm{ES}_\alpha\big(\Pi^k \# \mathbb{P}_r\big)\Big)$$

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- Mathematically,

$$D^* \in \arg\min_D \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{\boldsymbol{p} \sim \mathbb{P}_r} \left[ S_\alpha \left( D(\Pi^k \# \mathbb{P}_r); \boldsymbol{p} \right) \right],$$

# The generator

- Input: random noise $\boldsymbol{z} \sim \mathbb{P}_z$ distribution on the latent space
- Output $G(\boldsymbol{z})$: simulated price scenario
- Mapping parameterized by neural network:

$$G(\boldsymbol{z}; \boldsymbol{W}, \boldsymbol{b}) = \boldsymbol{W}_L \cdot \sigma\left(\boldsymbol{W}_{L-1} \cdots \sigma(\boldsymbol{W}_1 \boldsymbol{z} + \boldsymbol{b}_1)\right) + \boldsymbol{b}_L,$$

where

  - $\boldsymbol{W} = (\boldsymbol{W}_1, \boldsymbol{W}_2, \cdots, \boldsymbol{W}_L)$ and $\boldsymbol{b} = (\boldsymbol{b}_1, \boldsymbol{b}_2, \cdots, \boldsymbol{b}_L)$ represent the trainable parameters in the neural network, with $\boldsymbol{W}_l \in \mathbb{R}^{n_l \times n_{l-1}}$, $\boldsymbol{b}_l \in \mathbb{R}^{n_l \times 1}$ for $l = 1, 2, \cdots, L$
  - The operator $\sigma(\cdot)$ is a component-wise non-linear operator

# Objective of the generator

The goal of the generator is to best match the tail risks across the set of $K$ benchmark strategies:

$$\overbrace{D^*(\Pi^k \# \mathbb{P}_G) \approx \underbrace{D^*(\Pi^k \# \mathbb{P}_r) \approx \Big(\mathrm{VaR}_\alpha\big(\Pi^k \# \mathbb{P}_r\big), \mathrm{ES}_\alpha\big(\Pi^k \# \mathbb{P}_r\big)\Big)}_{\text{Goal of the discriminator}}}^{\text{Goal of the generator}}$$

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Mathematically,

$$\min_G \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{\boldsymbol{p} \sim \mathbb{P}_r} \left[ S_\alpha \Big( \underbrace{D^*(\Pi^k \# \mathbb{P}_G)}_{\text{VaR and ES prediction from } D^*} ; \ \Pi^k(\boldsymbol{p}) \Big) \right]$$

# Universal approximation of the generator

**Theorem (Cont, Cucuringu, X. and Zhang (2022))**

*Assume mild Lipschitz and regularity conditions. For any given error $\varepsilon > 0$, there exists a fully connected and feed-forward deep neural network $G_1$ with ReLU activation such that*

$$\left|\mathrm{VaR}_\alpha\Big(\Pi^k\#\big((\nabla G_1)\#\mathbb{P}_z\big)\Big) - \mathrm{VaR}_\alpha\Big(\Pi^k\#\mathbb{P}_r\Big)\right| < \varepsilon.$$

*for all $1 \le k \le K$. Here*

- *The depth of $G_1$: $L = \lceil \log n_1 \rceil$*
- *The width of $G_1$: $N = 2^L = 2^{\lceil \log n_1 \rceil}$*
- $n_1 = \mathcal{O}(\varepsilon^{-2})$

# Universal approximation of the generator

**Theorem (Cont, Cucuringu, X. and Zhang (2022))**

*Assume mild Lipschitz and regularity conditions ($\mathbb{P}_r$ has bounded $\beta$th moment ($1 < \beta \leq 2$).) For any given error $\varepsilon > 0$, there exists a fully connected and feed-forward deep neural network $G_2$ with ReLU activation such that*

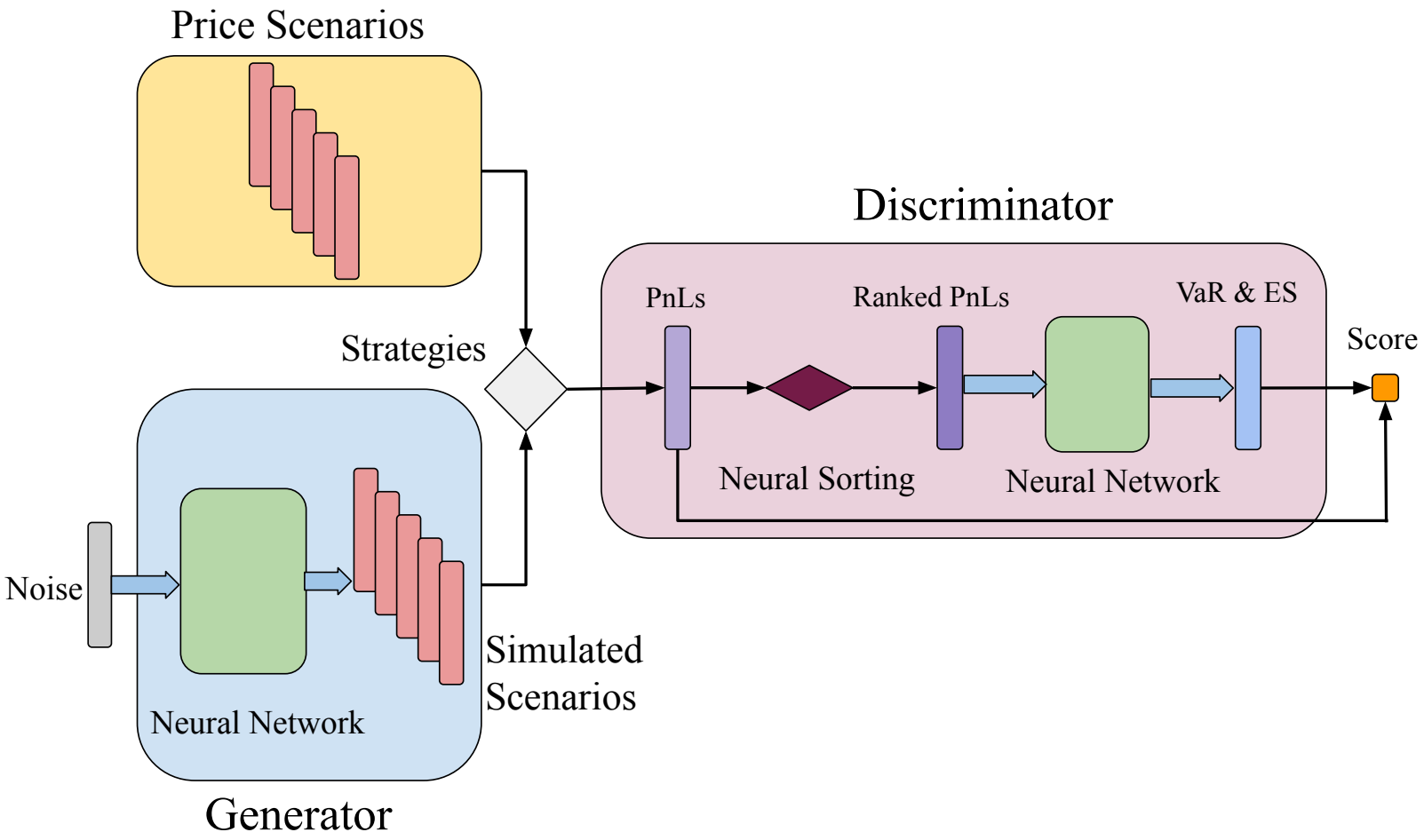
$$\left|\mathrm{ES}_\alpha\Big(\Pi^k\#((\nabla G_2)\#\mathbb{P}_z)\Big) - \mathrm{ES}_\alpha\Big(\Pi^k\#\mathbb{P}_r\Big)\right| < \varepsilon.$$

*for all $1 \leq k \leq K$. Here*

- *The depth of $G_2$: $L = \lceil \log n_2 \rceil$*
- *The width of $G_2$: $N = 2^L = 2^{\lceil \log n_2 \rceil}$*
- $n_2 = \mathcal{O}(\varepsilon^{-\frac{\beta}{\beta-1}})$

- Similar results also hold for risk measures that are Hölder continuous under the Wassterstein measure

# Summary of the architecture



# From bi-level optimization to max-min game

Our goal is to find a generator $G^*$ and a discriminator $D^*$ via the following bi-level optimization problem

$$G^* \in \arg\min_G \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{\boldsymbol{p}\sim\mathbb{P}_r} \left[ S_\alpha \Big( D^* \big( \Pi^k \# \mathbb{P}_G \big),\ \Pi^k(\boldsymbol{p}) \Big) \right],$$

where

$$D^* \in \arg\min_D \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{\boldsymbol{p}\sim\mathbb{P}_r} \left[ S_\alpha \Big( D\big( \Pi^k \# \mathbb{P}_r \big),\ \Pi^k(\boldsymbol{p}) \Big) \right].$$

$\hookrightarrow$ constrained optimization problem is usually difficult to solve in practice!

# From bi-level optimization to max-min game

**Theorem (Cont, Cucuringu, X. and Zhang (2022))**

*Under mild conditions, the bi-level optimization problem is equivalent to the following max-min game:*

$$\max_{D} \min_{G} \frac{1}{K} \sum_{k=1}^{K} \Bigg[ \mathbb{E}_{\boldsymbol{p} \sim \mathbb{P}_r} \Big[ S_\alpha \Big( D\big(\Pi^k \# \mathbb{P}_G\big),\ \Pi^k(\boldsymbol{p}) \Big) \Big] - \lambda\, \mathbb{E}_{\boldsymbol{p} \sim \mathbb{P}_r} \Big[ S_\alpha \Big( D\big(\Pi^k \# \mathbb{P}_r\big),\ \Pi^k(\boldsymbol{p}) \Big) \Big] \Bigg]$$

*with any Lagrange multiplier $\lambda > 0$.*

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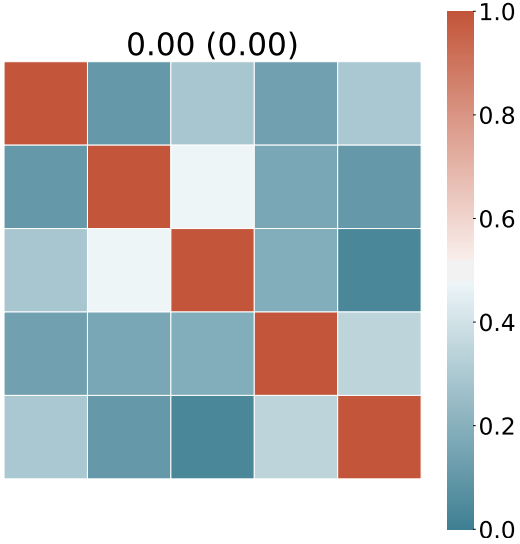
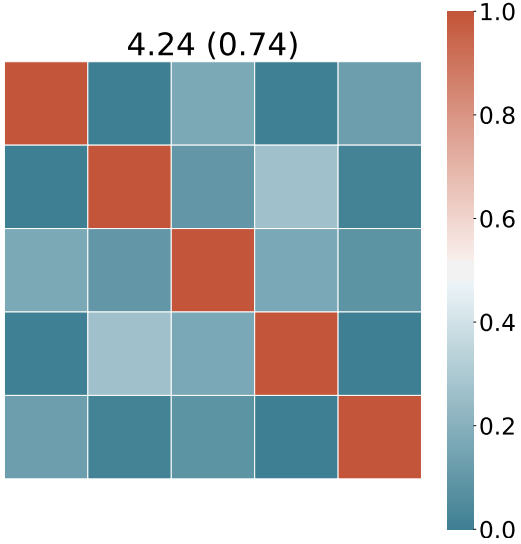
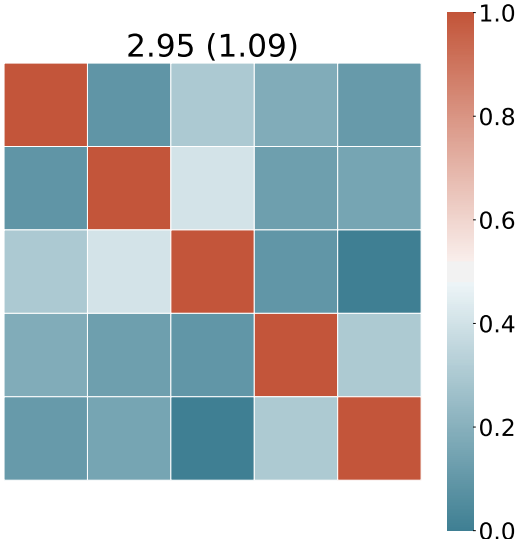
# Correlation patterns



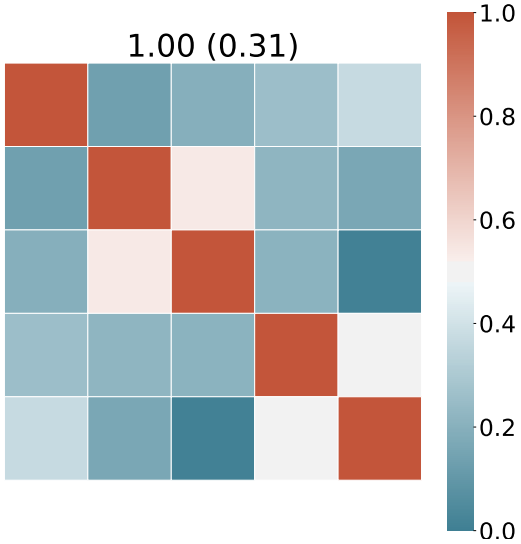

(c) Market data.




(d) TAIL-GAN-Raw.

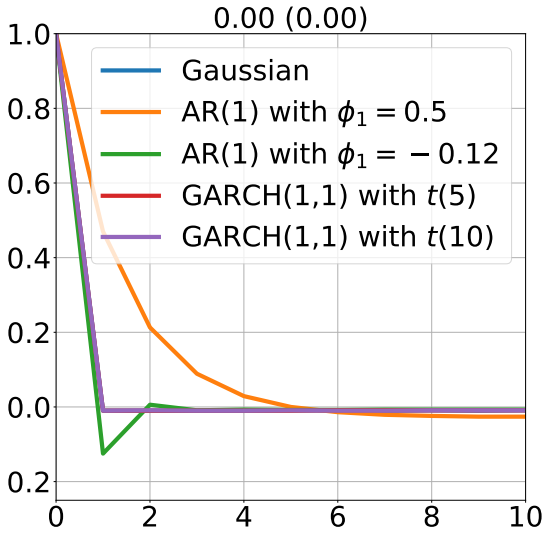



(e) TAIL-GAN-Static.

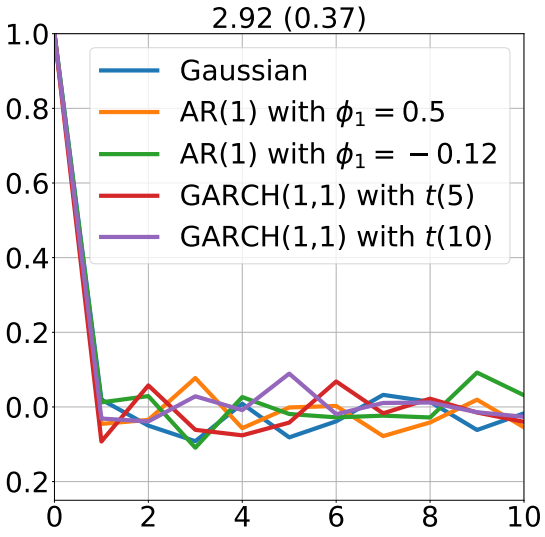



(f) TAIL-GAN.

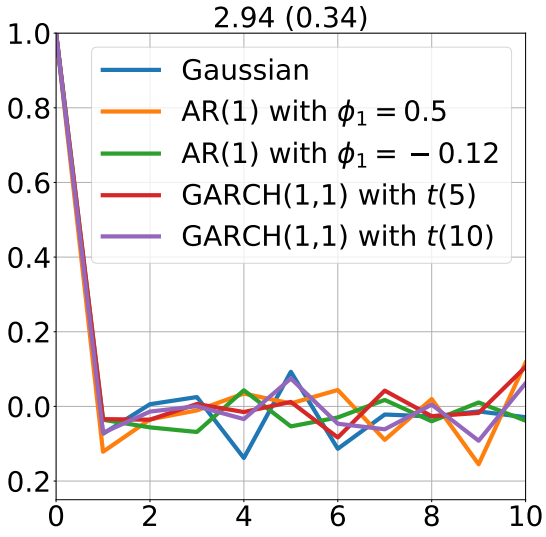
# Temporal patterns



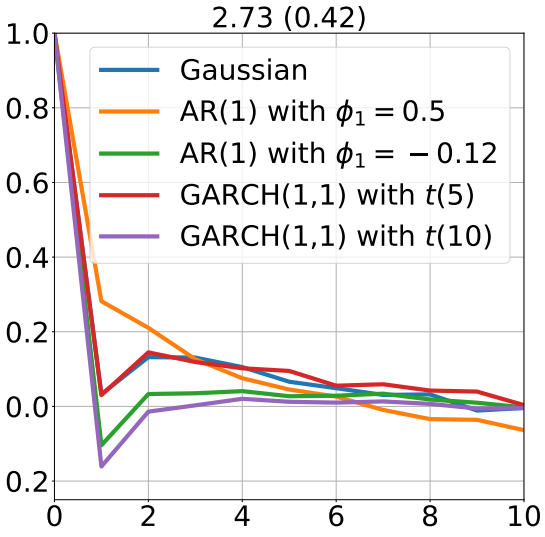

(a) Market data.




(b) TAIL-GAN-Raw.




(c) TAIL-GAN-Static.




(d) TAIL-GAN.

# Performance: out-of-sample error

| | "Oracle" | TAIL-GAN-Raw | TAIL-GAN-Static |
|---|---|---|---|
| % | 3.0 | 83.3 | 86.7 |
| (std) | (2.2) | (3.0) | (2.5) |
| | TAIL-GAN | WGAN | |
| | 4.6 | 21.3 | |
| | (1.6) | (2.2) | |

# Thank you!

https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3812973

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# Which score function to use?

In general, the score function is not unique...

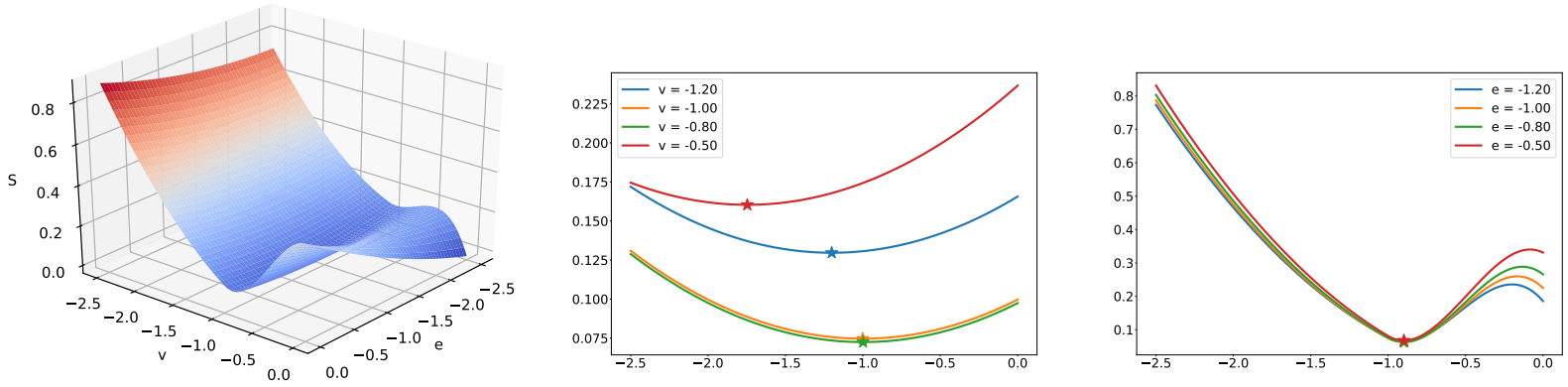
$$\begin{aligned} S_\alpha(v,e,x) &= (\mathbb{1}_{\{x\le v\}}-\alpha)(H_1(v)-H_1(x)) \\ &\quad + \frac{1}{\alpha}H_2'(e)\mathbb{1}_{\{x\le v\}}(v-x) + H_2'(e)(e-v) - H_2(e) \end{aligned}$$

- [Fissler et al., 2016] proposes:
  $H_2:\mathbb{R}\to\mathbb{R}$ strictly convex and $R_\alpha(v,e) := \frac{1}{\alpha}vH_2'(e)+H_1(v)$ strictly increasing in $v$
- [Acerbi and Szekely, 2014] propose:
  $H_1(v) = -\frac{W_\alpha}{2}v^2,\ H_2(e) = \frac{\alpha}{2}e^2, \quad \text{with} \quad \frac{\mathrm{ES}_\alpha(\mu)}{\mathrm{VaR}_\alpha(\mu)} \ge W_\alpha \ge 1$

# Which score function to use?

**Proposition (Cont, Cucuringu, X. and Zhang (2022))**

*The score function proposed in [Acerbi and Szekely, 2014] has a better optimization landscape near the global minimum.*




(a) $s_\alpha(v, e)$.

(b) $s_\alpha(v, e)$ as a function of $e$.

(c) $s_\alpha(v, e)$ as a function of $v$.

Figure 3: Landscape of $s_\alpha(v, e) = \mathbb{E}_{X \sim U([-1,1])}[S_\alpha(v, e, X)]$ based on [Acerbi and Szekely, 2014]